

AN AUTOMATED SYSML-BASED SURROGATE MODEL CONSTRUCTION APPROACH FOR PRODUCT CAD

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ABSTRACT

An automated method for constructing proxy models of product CAD models based on the standardized SysML language is proposed in this study. The method achieves the abstraction and standardized representation of product CAD models by defining the geometric features and physical properties of part CAD models, the dependencies of logical relationships and constraints between different parts, as well as the assembly CAD models representing the relative positions and connections between multiple parts. Transformation rules for mapping these elements to SysML model elements are established. The automated transformation from native product CAD models to SysML-based proxy models is realized using ATL, enabling the full-process automated construction of system-level proxy models. Using a robotic arm product CAD model as an example, the proxy model is constructed, demonstrating the abstraction and representation process of the product CAD model and validating the feasibility of the proposed approach. This method integrates product CAD models into the digital thread in the form of system proxy models, laying the foundation for promoting the rapid transmission and feedback of information throughout the product lifecycle and facilitating cross-domain integration and collaboration.

Keywords: Digital Thread, PLM, SysML Language, Native Model Transformation

1. INTRODUCTION

With the rapid development of the digital economy, companies are accelerating their digital transformation[1]. However, during the transformation process, many enterprises encounter the issue of information silo. Particularly, there is a severe data gap between business systems, leading to data inconsistency, incompleteness, and untimeliness[2]. For example, design modifications are not timely fed back to the manufacturing and operation stages, leading to the inability to smoothly form a closed-loop between forward design and negative feedback. As a result, this “broken chain” of

information makes it difficult to identify and resolve issues in the production process in a timely manner, thereby affecting product quality and production efficiency[3-5].

Native models refer to detailed data models used across various stages of the product lifecycle, such as CAD models and process models. While native models serve as the core of product design and contain extensive technical details, they have notable limitations in supporting cross-departmental and cross-functional collaboration. Specifically, the complex geometric data and manufacturing process information generated during the design phase are often difficult to directly apply to the production or operation and maintenance phases. This leads to the formation of information silos and reduces the speed and accuracy of design change responses. Therefore, relying solely on the integration of native models cannot meet the demands for cross-domain collaboration in digital transformation.

The Digital Thread[6], as an emerging technological framework, integrates data from various stages of design, manufacturing, and operation and maintenance, creating a unified data flow that spans the entire product lifecycle[7]. By introducing surrogate models, key information from the native models is simplified and abstracted, removing redundant details while retaining critical data[8]. This promotes cross-departmental and cross-functional collaboration, driving the standardization and unification of information, thereby optimizing processes at each stage and enhancing collaborative efficiency.

Systems Modeling Language (SysML), as a standardized modeling language, helps eliminate discrepancies in expression and terminology across different modeling languages, supporting information abstraction and business collaboration[9]. It serves as an effective means for expressing surrogate models within the Digital Thread framework. With Digital Thread technology, key information can be rapidly captured and transmitted, improving design response speed and accuracy, while also ensuring the timely implementation of feedback mechanisms.

Product CAD models, as a crucial component of native models, play a vital role in product design, manufacturing, and operation. They provide precise geometric data for product manufacturing and lay the foundation for subsequent simulation analysis, process planning, and quality control[10, 11]. However, in traditional PLM systems, the integration between CAD models and other business systems is limited, especially in terms of cross-departmental collaboration and information sharing. By constructing surrogate models of product CAD models, abstracting their key information, and integrating them into the digital thread, these limitations can be effectively addressed. Surrogate models simplify the native CAD model while retaining critical information, thereby facilitating the rapid flow and sharing of data across different stages and systems, enhancing collaboration efficiency across various departments within the enterprise.

This paper proposes an automated construction method for surrogate models of product CAD models. By analyzing the key model features and business requirements of product CAD models, mapping rules for abstracting model characteristics and attributes are constructed, retaining critical product CAD model information. This approach achieves a semantically standardized SysML-based system-level representation of the product CAD model, enabling the system-level automatic integration of design information into the digital thread. It also provides the foundation for interaction with system-level surrogate models of manufacturing and operation information. The structure of the paper is as follows: Section 2 introduces the concepts of native product models and SysML-based surrogate models. Section 3 provides a detailed description of the mapping rules for parts, dependencies, and assemblies from product CAD models to surrogate models. Section 4 primarily presents an automated transformation process for converting product CAD models into surrogate models. Finally, the paper concludes with a summary and outlines potential directions for future research.

2. RELATED WORKS

2.1 Native model

Native models refer to the comprehensive data representations employed throughout the entire product lifecycle, encapsulating detailed technical information and design parameters. These models typically encompass core data required for product design, manufacturing, and maintenance, including CAD models, manufacturing process models, simulation models, and quality inspection models. Serving as the central reference for design activities, native models capture a complete spectrum of information related to geometric structures, material properties, manufacturing processes, performance requirements, and operational conditions, thus providing accurate and reliable data support to various business functions.

Based on the distinct requirements of each lifecycle phase, native models can be broadly classified into three primary categories. First, design-stage native models comprise CAD models and simulation models. CAD models support product conceptualization and visualization, documenting product

structure, component relationships, and dimensional tolerances; simulation models address aspects such as stress-strain analysis, fluid flow, or thermodynamics, thereby offering insights for design optimization and decision-making. Second, manufacturing-stage native models include process models and fixture/jig models. Process models specify machining paths, tooling choices, assembly procedures, and process parameters to guide shop-floor operations, while fixture/jig models incorporate design and simulation to ensure precision in positioning and operational stability. Finally, operation and maintenance-stage native models encompass maintenance models and quality traceability models: maintenance models provide guidance for monitoring, fault diagnosis, and upkeep activities once the product is in service; quality traceability models record production batches, inspection results, and component replacements, enabling comprehensive lifecycle tracking and facilitating continuous improvement.

Product CAD models serve as comprehensive digital representations that precisely capture a product's geometry, structure, assembly relationships, and physical properties[12-14]. In contrast to traditional two-dimensional drawings, CAD models not only depict static geometric features but also integrate material information, dimensional parameters, manufacturing tolerances, and inter-component relationships. This level of integration enables designers to visualize and refine product configurations within a virtual environment, thereby minimizing the risk of design flaws or suboptimal solutions.

Throughout the product lifecycle, CAD models provide critical support at every stage. During the design phase, they furnish detailed data for rapid iterations and optimizations. In the manufacturing phase, the dimensional specifications and process requirements embedded in CAD models ensure precise production. During assembly, CAD models facilitate accurate alignment and integration of components, while in the after-sales service phase, they offer reliable references for maintenance, modification, and upgrades. By spanning design, manufacturing, assembly, and service, CAD models enable seamless data exchange and collaborative decision-making, thereby enhancing the overall efficiency and quality of product development[15].

2.2 Surrogate model based on SysML

SysML as a standardized graphical modeling language, is primarily applied to multidisciplinary system design[16]. Unlike traditional CAD models that focus on geometric design, SysML emphasizes the relationships between system structure, behavior, and requirements while providing comprehensive descriptions across multiple levels. It supports system description and analysis by defining systems engineering-specific elements such as requirements, functions, components, and interfaces, and introduces dedicated diagram types like requirement diagrams and block definition diagrams tailored for systems engineering.

The SysML-based surrogate model is a simplified modeling approach aimed at enabling system-level design and optimization[17]. By streamlining and abstracting critical information from product CAD models—eliminating redundant data and expressing system functionality, performance, and

structure in a standardized manner—it transforms complex CAD models into concise, structured representations. This conversion not only enhances model usability for cross-disciplinary and cross-tool collaboration but also supports modeling and analysis needs across diverse domains, reducing system design complexity and information errors[18]. Consequently, SysML-based surrogate models significantly improve system-level optimization and smart manufacturing capabilities for complex products.

3. OVERVIEW OF METHODOLOGY

To integrate product CAD models into a digital-thread-based collaborative management framework across the entire product lifecycle, this paper proposes an automated SysML-based surrogate model construction approach for product CAD. (see in FIGURE 1). The methodology centers on two main components.

First, mapping rules are established to correlate key elements of the native CAD model—such as part attributes, dependency relationships, and assembly constraints—with SysML constructs (e.g., Blocks, Properties, and Constraints). These rules specify how to extract and represent essential product information, including geometric dimensions, physical properties, and inter-part interfaces.

Second, an automatic construction process applies these mapping rules to generate the SysML-based surrogate model. In this process, the relevant information is extracted from the original CAD files into a standardized XML format, which the Atlas Transformation Language (ATL) then uses to translate into SysML elements. This translation preserves critical design details while filtering out superfluous geometry, thereby improving model manageability and cross-domain clarity. The resulting SysML surrogate model can be directly imported into

systems engineering tools, providing a streamlined basis for system-level analysis and collaboration. By structuring the methodology around clear mapping rules and automated model transformation, the proposed approach ensures accurate, consistent, and efficient integration of product CAD data into broader digital thread workflows.

4. MODEL TRANSFORMATION MAPPING RULES

Three types of model transformation mapping rules are defined in this section, including part mapping rules, part mapping dependencies, and assembly model mapping rules.

4.1 Part mapping

In multidisciplinary collaboration and cross-domain integrated system design, the surrogate models of parts are simplified through formal expressions, focusing on key geometric parameters and physical characteristics to support system-level design, optimization, and analysis.

The geometric information of parts is the core of product CAD models, defining the shape and dimensions of the product. The CAD model utilizes various geometric features (such as holes, slots, chamfers, and curves) to describe the shape and structure of the part. Under the SysML-based surrogate model framework, these complex geometric shapes are formalized by simplifying them into fundamental parameters (such as length, width, height, etc.) for the key geometric features.

The physical properties of parts, including materials, mass, and inertia matrix, not only affect the manufacturing and performance of the product but also directly influence its functional performance in system-level design. By using SysML, new constructs can be created through property field storage, thereby formalizing the expression of physical properties.

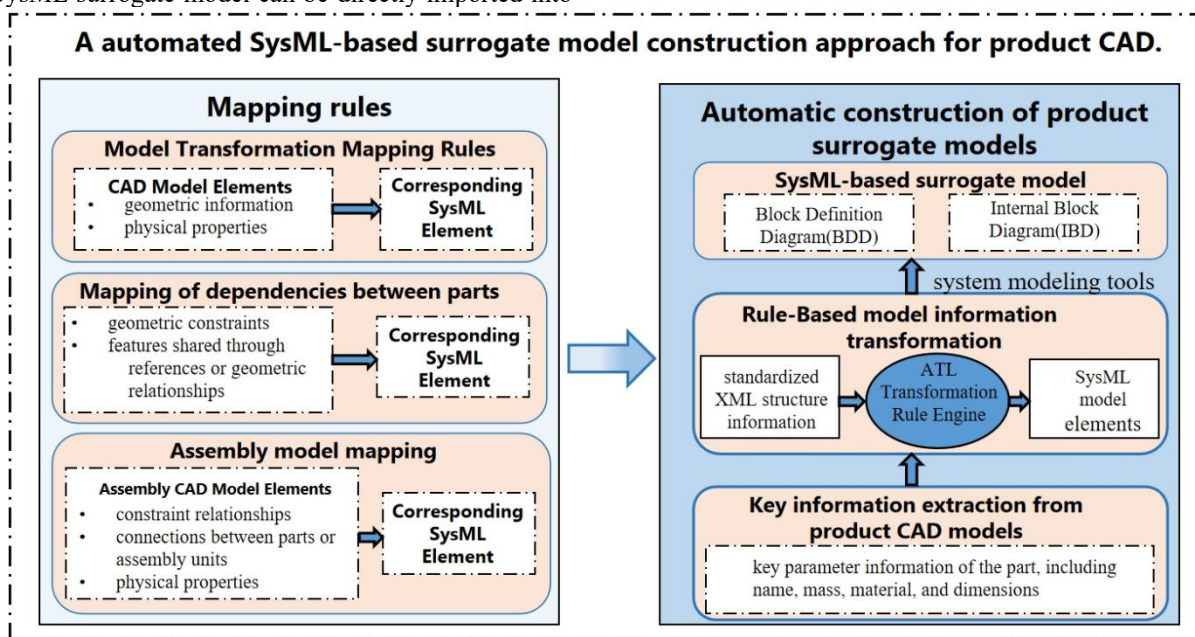


FIGURE 1: AN AUTOMATED SYSML-BASED SURROGATE MODEL CONSTRUCTION APPROACH FOR PRODUCT CAD

Geometric information and physical properties of parts constitute essential data within the systems engineering process; however, the specific information required varies subtly depending on particular systems engineering decisions or simulation tasks. This study defines mapping rules for critical geometric and physical property information, allowing the extension of these rules to accommodate the detailed information demands of specific tasks. Mapping relationships are determined by considering each element's role and information representation within both the native CAD model and the SysML model. The TABLE 1 below outlines how geometric and physical attributes of parts in the CAD model correspond to SysML constructs. Specifically, the Part element in the CAD model is mapped to the Block element in SysML, as both are used to describe and organize complex systems or products in a modular manner, serving as fundamental building blocks. Similarly, the Annotation element in the CAD model corresponds to the Comment element in SysML, given that both provide auxiliary textual information to clarify or explain model details. For physical attributes such as Material and Mass, SysML leverages the Property field to store and manage these characteristics, ensuring their effective transmission and utilization in system-level modeling.

TABLE 1: MAPPING OF PART CAD MODEL ELEMENTS TO SURROGATE MODEL ELEMENTS

CAD Model Elements	Description	Corresponding SysML Element
Part	Parts, the basic building units in a system	Block
Feature	Geometric features of parts, such as holes, bosses, etc.	Property
Geometric Constraint	Geometric constraints, used to restrict the geometric features of parts.	Constraint
Material	Materials, used to describe the physical properties of parts.	Property
Mass	Mass, used to describe the mass characteristics of parts.	Property
Dimension	Dimensions, used to describe the size characteristics of parts.	Values
Annotation	Annotations, used to describe additional information about parts.	Comment
Type	Types of parts, primarily divided into four categories: bushing, cover plate, fork frame, and casing.	Property
Roughness	Roughness, the small spacing and tiny peaks and valleys of the machined surface's irregularity.	Property

4.2 Mapping of dependencies between parts

The dependencies between parts describe the logical relationships and constraints among different parts, involving the

fit between parts, reference geometries, shared geometric features, or relative motion. These relationships define how different parts in a product work together to perform specific functions and are crucial for system design, analysis, and optimization in cross-domain integration and collaboration.

The dependencies between parts include geometric constraints (such as alignment, parallelism, and coaxiality) and features shared through references or geometric relationships, ensuring the structural integrity and functional correctness of the product. In complex products, the interdependencies among multiple parts determine the performance and functional behavior of the product.

By abstracting the dependencies between parts into interfaces, surrogate models are used to represent these relationships. In the surrogate model, the interface defines indirect interactions between parts through shared or reference geometries, such as coaxiality. For example, the interface in a surrogate model ensures that the corresponding logical relationships and constraints between two parts, as represented in the CAD model, are preserved.

In the surrogate model for dependencies, the dependent part is called the child component, and the part providing the dependency information is called the parent component. The parent component explicitly declares the elements available for use by other parts through the <<CADPublication>> interface, while the child component expresses its dependency by referencing these published elements. It is noteworthy that the parent-child relationships between components with dependencies do not necessarily align with the model assembly hierarchy. In surrogate model construction, components that explicitly declare interfaces for publishing information are defined as parent components, while those referencing the published elements are defined as child components. This publishing mechanism renders the geometry replacement process more transparent and streamlined, ensuring that changes in parent components propagate accordingly to child components. For example, the bottom shape of the slider component (child component) depends on the dimensional and geometric profile of the guide rail in the base component (parent component). The dependency is established through the sketch dimension and shape elements published by the guide rail, thereby achieving accurate preservation and standardized description of the dependency relationship.

4.3 Assembly model mapping

The assembly of part instances is called a product, and the assembly relationships are one of the core pieces of information in the product's CAD model. These relationships define the relative positions and connections between multiple parts within the product. Correct assembly relationships ensure the operability and stability of the product during actual assembly. In the construction of surrogate models for assembly CAD models, the focus is on the constraint relationships and connections between parts or assembly units, as well as the decomposition of parts in the assembly surrogate model. This approach enables the full utilization of key information from the

assembly surrogate model in design, manufacturing, and other services throughout the product's lifecycle analysis and optimization process. The detailed mapping of the key model elements from the assembly CAD model to the SysML model elements is shown in the TABLE 2.

In this context, the Product in the assembly CAD model and the Package in SysML both serve the purpose of organizing and managing complex structures. The Product in the assembly CAD model is a complex structure composed of multiple parts, components, or assemblies, while the Package in SysML is a container for organizing and managing related model elements. Therefore, a mapping relationship is established between the two. Constraints, Assembly Constraints, Motion Constraints, and Annotations within the assembly CAD model represent the constraint relationships between parts and assemblies. Therefore, establishing mapping relationships between these CAD model elements and the Constraint model elements in SysML is essential. Different types of SysML Constraints that reflect various motion and assembly restrictions can be distinguished during model construction by adding specially defined attribute tags.

TABLE 2: MAPPING OF ASSEMBLY CAD MODEL ELEMENTS TO SURROGATE MODEL ELEMENTS

Assembly CAD Model Elements	Description	Corresponding SysML Elements
Product	Part Assembly, used to organize and manage parts	Package
Constraint	Connection relationships, used to describe the relative position and motion relationships between parts	Constraint
Assbly Constraint	Assembly constraints, used to describe the assembly relationships between parts	Constraint
Motion Constraint	Motion constraints, used to describe the motion relationships between parts	Constraint
Annotion	Annotations, used to describe additional information about parts	Constraint
Material Property	Material properties, used to define the physical properties of materials	Property
Physical Property	Physical properties, used to describe the physical characteristics of parts	Property

5. AUTOMATIC CONSTRUCTION OF PRODUCT SURROGATE MODELS

5.1 Key information extraction from product CAD models

The product CAD model contains a large amount of detailed information regarding geometric shapes, physical properties, and assembly relationships. In order to integrate this information into the SysML-based system-level surrogate model, a multi-level extraction process is employed to structure these key pieces of information while abstracting unnecessary details for efficient representation within the system model.

The information extraction process from the product's CAD model is analyzed and processed from two aspects: part information and assembly information. The extraction of part information primarily focuses on the detailed description of individual components, such as dimensions, materials, and other properties, while the extraction of assembly information emphasizes the relationships between multiple parts and the constraint relationships between assembly units.

The part information extraction process is shown in the FIGURE 2. First, the part model is prepared. Next, the system locates and accesses the model's core data, ensuring proper identification of pertinent geometric, physical, and assembly information. Subsequently, automated algorithms extract the necessary parameter details, converting them into a standardized format for further analysis. Finally, the parameter information table is generated to support downstream activities such as surrogate model construction or system-level design evaluations.

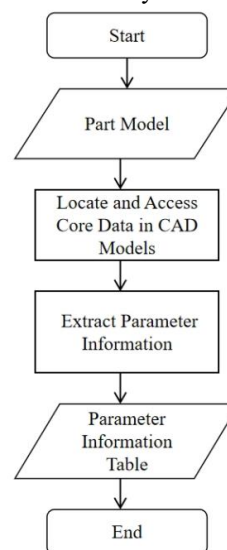


FIGURE 2: PART INFORMATION EXTRACTION PROCESS

The assembly information extraction process is shown in the FIGURE 3. The model document pointer of the assembly's CAD model is obtained, and the parameter information of the parts under the root directory node is sequentially read. It is then determined whether the multi-part assembly constitutes an assembly unit. The process continues recursively until the information of the lowest-level assembly unit is obtained. Finally, the information is output as a standard format assembly parameter information table and converted into the corresponding XML file.

5.2 Rule-Based model information transformation

Atlas Transformation Language is a rule-based language specifically designed for model transformation. By defining the mapping rules between source models and target models, it enables the automatic transformation of complex models. The model information transformation process based on ATL is illustrated as follows in the FIGURE 4. The ATL tools first load the source model (standardized XML structure information extracted from the product CAD model) and parse it. Based on the mapping rules defined in Section 2, the ATL engine matches each element from the source model with the corresponding element in the target model. The target model can then be directly parsed and used by system modeling tools (such as MagicDraw), automatically constructing the SysML-based surrogate model,

which can be used for subsequent system-level design analysis and optimization.

Section 2 discusses the mapping relationship between the product CAD model and the SysML model, specifically how elements such as parts, geometric features, attributes, and assembly constraints from the product CAD model are mapped to Block modules, properties, and constraints in the SysML-based surrogate model. This allows the details of the product CAD model to be abstractly represented at the system design level, providing support for systems engineering. These mapping relationships serve as the foundation for implementing the ATL-based transformation rules, as shown in the FIGURE 5, which abstractly describes the mapping from the product CAD model to the surrogate model.

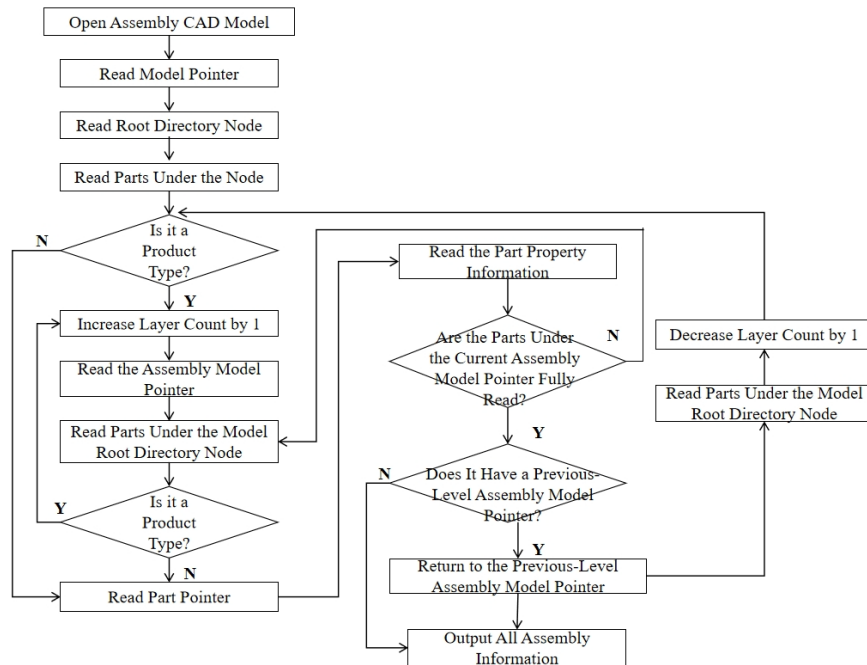


FIGURE 3: ASSEMBLY INFORMATION EXTRACTION PROCESS

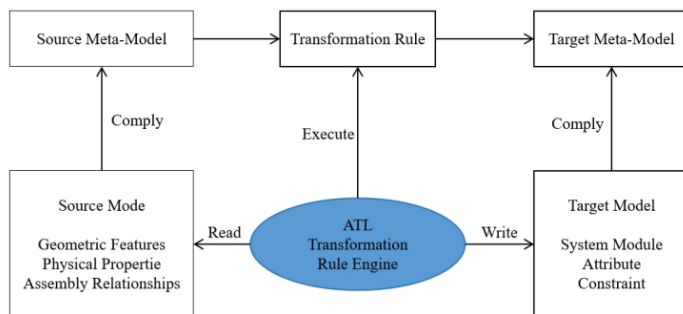


FIGURE 4: PRINCIPLE OF MODEL INFORMATION TRANSFORMATION BASED ON ATL

```
// Rule: Part to SysML Block
rule PartToBlock {
  from
    p : CAD::Part // Source model: Part from CAD model
  to
    b : SysML::Block // Target model: Block in SysML model
  do
    b.name := p.name; // Map part name to Block name
    b.description := p.description; // Map part description to Block description
    b.type := p.type; // Map part type to Block type
  }

// Rule: Feature to SysML Property
rule FeatureToProperty {
  from
    f : CAD::Feature // Source model: Feature from CAD model
  to
    pr : SysML::Property // Target model: Property in SysML model
  do
    pr.name := f.name; // Map feature name to Property name
    pr.type := 'Feature'; // Set the property type as 'Feature'
    pr.value := f.value; // Map feature value to property value
  }
}
```

FIGURE 5: ATL MAPPING RULES CODE EXAMPLE

6. EXAMPLE OF SURROGATE MODEL CONSTRUCTION FOR ROBOTIC ARM CAD MODEL

The transformation from product CAD models to SysML-based surrogate models helps capture key product information for system-level design and optimization. The following presents a robotic arm product CAD model as an example to demonstrate the process of converting a product CAD model into a SysML-based surrogate model, validating the technical feasibility of the proposed method.

The robotic arm product consists of the base part, arm 1, arm 2, joint 1, and joint 2, with the base, arm 1, and arm 2 connected through joints 1 and 2. The specific geometric parameters of the robotic arm are shown in the TABLE 3, and its CAD model is depicted in the FIGURE 6.

TABLE 3: ROBOTIC ARM PART GEOMETRIC PARAMETERS

Part Type	Part Parameters (mm)
Arm	Length=80,R=20,Width=40,Height=10
Joint	R=10,Height=10
Base	R=10,Depth=10

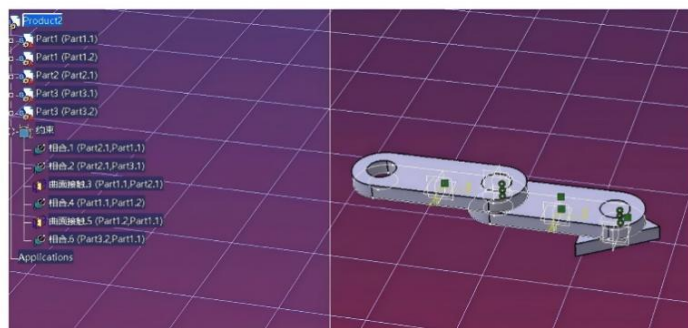


FIGURE 6: SCHEMATIC DIAGRAM OF ROBOTIC ARM PRODUCT CAD MODEL

Based on the model information extraction method described in Section 3.1, key parameter information of the robotic arm STEP file is automatically extracted using a Python script, generating the corresponding parts' key parameter information table as shown in the TABLE 4. The robotic arm constraint information, extracted according to the constraint data of geometric features, is presented in the TABLE 5.

TABLE 4: KEY PARAMETER INFORMATION OF ROBOTIC ARM PARTS

Key Parameter Name	Key Parameter Value
'arm length'	80
'arm width'	40

'arm thickness'	10
'arm hole diameter'	10
'outer radius'	20
'base hole diameter'	10
'density'	2710
'joint radius'	10
'joint thickness'	20

TABLE 5: ROBOTIC ARM CONSTRAINT INFORMATION

Constraint	Part 1	Part 2
'Constraint\Coincidence.1\Coincidence.1'	joint2	arm1
'Constraint\Coincidence.2\Coincidence.2'	base	arm1
'Constraint\Surface Contact.3\Surface Contact.3'	base	arm1
'Constraint\Surface Contact.4\Surface Contact.4'	Joint2	arm1
'Constraint\Coincidence.5\Coincidence.5'	arm2	arm1
'Constraint\Surface Contact.6\Surface Contact.6'	arm1	arm2
'Constraint\Coincidence.7\Coincidence.7'	joint1	arm2

After reading the key parameter and constraint information from the robotic arm CAD model, an XML document containing the corresponding specific geometric parameter information is generated, as shown in the FIGURE 7.

By constructing a surrogate model transformation rule library, CAD product model information is converted into a standardized XML file, which is then transformed into a SysML-compatible XML document that can be parsed by system engineering modeling software using the ATL converter. This process generates the corresponding product surrogate model. The SysML-based surrogate model representation of the robotic arm is shown in the FIGURE 8 and FIGURE 9. The block definition diagram primarily describes the specific geometric parameters and density information of each part of the robotic arm, while the internal block diagram outlines the constraint relationships between the various geometric features.

Through the construction of a simplified, abstracted system-level surrogate model for the robotic arm product CAD model, key geometric, constraint, and other relevant information from the product CAD model is retained. This enables the creation of system-level block definition and internal module diagrams, thereby validating the technical feasibility of the proposed method. The construction of the system surrogate model for the robotic arm product CAD model lays the foundation for supporting design analysis and optimization in manufacturing, operation, and other stages of its system, facilitating cross-domain information transfer and collaboration.

Key parameter information of the arm

```

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  <ownedAttribute xmi:type="uml:Property" xmi:id="_19_0_2_8270274_1716272672079_195039_42603" name="Length" aggregation="composite" type="_19_1" />
  <ownedAttribute xmi:type="uml:Property" xmi:id="_19_0_2_8270274_1716272672079_195039_42603" name="depth" aggregation="composite" type="_19_1" />
  <ownedAttribute xmi:type="uml:Property" xmi:id="_19_0_2_8270274_1716272672079_250829_42604" name="width" aggregation="composite" type="_19_1" />
  <ownedAttribute xmi:type="uml:Property" xmi:id="_19_0_2_8270274_1716272672079_681831_42605" name="round_diameter" aggregation="composite" type="_19_1" />
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<packagedElement xmi:type="uml:DataType" xmi:id="_19_0_2_8270274_1716272672079_406117_42607" name="20mm"/>

```

Component Interfaces and Interactions

Data

FIGURE 7: EXAMPLE OF CONVERTING ROBOTIC ARM MODEL INFORMATION TO STANDARD XML FILE

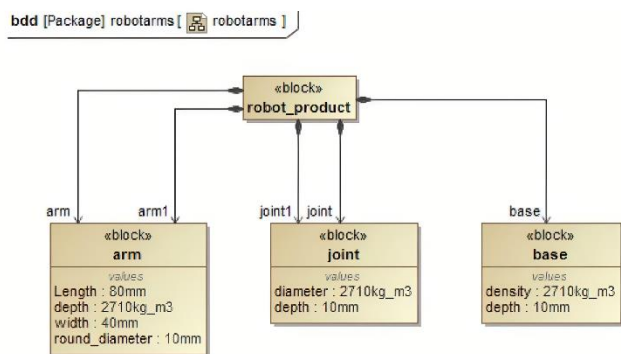


FIGURE 8: ROBOTIC ARM SURROGATE MODEL BLOCK DEFINITION DIAGRAM

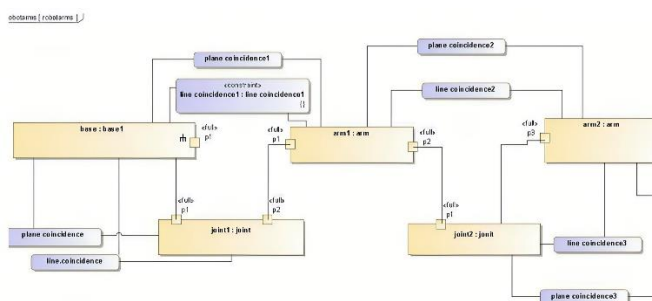


FIGURE 9: ROBOTIC ARM SURROGATE MODEL INTERNAL BLOCK DIAGRAM

7. CONCLUSION

This study proposes an automated approach for constructing surrogate models from product CAD models. Specifically, we propose a set of transformation mapping rules that convert product CAD models into SysML-based surrogate models. The method describes, in detail, the mapping and transformation of part-level CAD models, inter-part dependencies, and assembly models. By extracting key information from the product CAD model into a standardized XML file, and applying these mapping

rules via the ATL transformation language, we achieve automated surrogate model construction.

The primary contributions of this work are as follows:

(1) The concept of a “native model” is introduced to emphasize comprehensive consideration of product lifecycle information—including geometry, physical properties, functionality, and behavior—in the early design phase.

(2) An automated construction method for SysML-based surrogate models from product CAD models is introduced, detailing the general rules for abstracting product CAD models into surrogate models. The method utilizes XML standard format files for the automated construction of surrogate models, laying the foundation for integrating native models into the digital thread, thereby enabling seamless data flow and collaboration throughout the entire product lifecycle.

(3) A practical example involving a robotic arm verifies the feasibility of the proposed surrogate model construction method.

Future research will focus on further developing automated surrogate model construction technologies, exploring mapping rules and standardized expressions for CAD model topological relationships and associated data, as well as deeply analyzing and representing design intent and kinematic features of product CAD models. This will involve refining mapping rules and promoting deeper system-level applications and developments throughout the product's entire lifecycle. At the same time, the automatic transformation of system-level models for product CAD models of varying complexity will be validated, continually improving the mapping rules from product CAD models to SysML-based surrogate models. Additionally, the study will investigate the data flow mechanisms and data reasoning decision methods between different surrogate models, optimizing the data collaboration and management mechanisms within the digital thread throughout the product lifecycle.

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REFERENCES

- [1] Williams, L. D., 2021, "Concepts of Digital Economy and Industry 4.0 in Intelligent and information systems," *International Journal of Intelligent Networks*, 2, pp. 122-129.
- [2] Li, J., Tao, F., Cheng, Y., and Zhao, L., 2015, "Big Data in product lifecycle management," *The International Journal of Advanced Manufacturing Technology*, 81(1), pp. 667-684.
- [3] Li, P., Cheng, Y., Song, W., and Tao, F., 2020, "Manufacturing services collaboration: connotation, framework, key technologies, and research issues," *The International Journal of Advanced Manufacturing Technology*, 110(9), pp. 2573-2589.
- [4] Mourtzis, D., 2020, "Simulation in the design and operation of manufacturing systems: state of the art and new trends," *International Journal of Production Research*, 58(7), pp. 1927-1949.
- [5] Sluga, A., Butala, P., and Peklenik, J., 2005, "A Conceptual Framework for Collaborative Design and Operations of Manufacturing Work Systems," *CIRP Annals*, 54(1), pp. 437-440.
- [6] Abdel-Aty, T. A., and Negri, E., 2024, "Conceptualizing the digital thread for smart manufacturing: a systematic literature review," *Journal of Intelligent Manufacturing*, 35(8), pp. 3629-3653.
- [7] Zhang, Q., Liu, J., and Chen, X., 2024, "A Literature Review of the Digital Thread: Definition, Key Technologies, and Applications," *Systems*.
- [8] Dubicki, S., and Gorospe, R., "Lessons Learned from Constructing a Digital Thread," *Proc. 2024 IEEE International Systems Conference (SysCon)*, pp. 1-3.
- [9] Bassi, L., Secchi, C., Bonfe, M., and Fantuzzi, C., 2011, "A SysML-Based Methodology for Manufacturing Machinery Modeling and Design," *IEEE/ASME Transactions on Mechatronics*, 16(6), pp. 1049-1062.
- [10] Hoffman, C. M., and Joan-Arinyo, R., 1998, "CAD and the product master model," *Computer-Aided Design*, 30(11), pp. 905-918.
- [11] Baba, Y., and Nobeoka, K., 1998, "Towards knowledge-based product development: the 3-D CAD model of knowledge creation," *Research Policy*, 26(6), pp. 643-659.
- [12] Bharadwaj, A. G., and Starly, B., 2022, "Knowledge graph construction for product designs from large CAD model repositories," *Advanced Engineering Informatics*, 53, p. 101680.
- [13] Brière-Côté, A., Rivest, L., and Maranzana, R., 2013, "3D CAD Model Comparison: An Evaluation of Model Difference Identification Technologies," *Computer-Aided Design and Applications*, 10(2), pp. 173-195.
- [14] Brière-Côté, A., Rivest, L., and Maranzana, R., 2012, "Comparing 3D CAD Models: Uses, Methods, Tools and Perspectives," *Computer-Aided Design and Applications*, 9(6), pp. 771-794.
- [15] Zhang, C., Dong, L., and Wang, Y., 2023, "Design-Manufacturing-Operation & Maintenance (O&M) Integration of Complex Product Based on Digital Twin," *Applied Sciences*.
- [16] Hampson, K., 2015, "Technical Evaluation of the Systems Modeling Language (SysML)," *Procedia Computer Science*, 44, pp. 403-412.
- [17] Baduel, R., Chami, M., Bruel, J.-M., and Ober, I., "SysML Models Verification and Validation in an Industrial Context: Challenges and Experimentation," *Proc. Modelling Foundations and Applications*, A. Pierantonio, and S. Trujillo, eds., Springer International Publishing, pp. 132-146.
- [18] Graves, H., and Bijan, Y., 2011, "Using formal methods with SysML in aerospace design and engineering," *Annals of Mathematics and Artificial Intelligence*, 63(1), pp. 53-102.